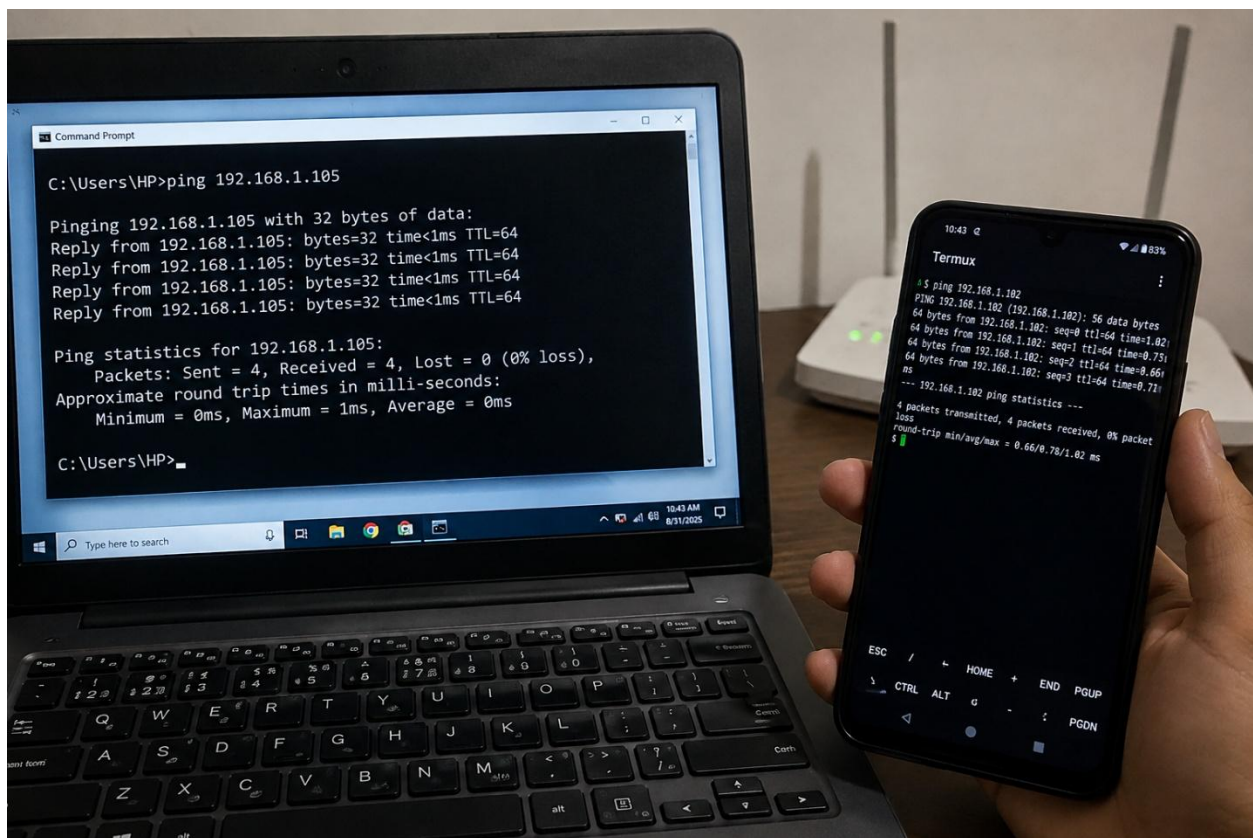




Assignment 5

Network Testing Practical

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.





Laptop (Windows)

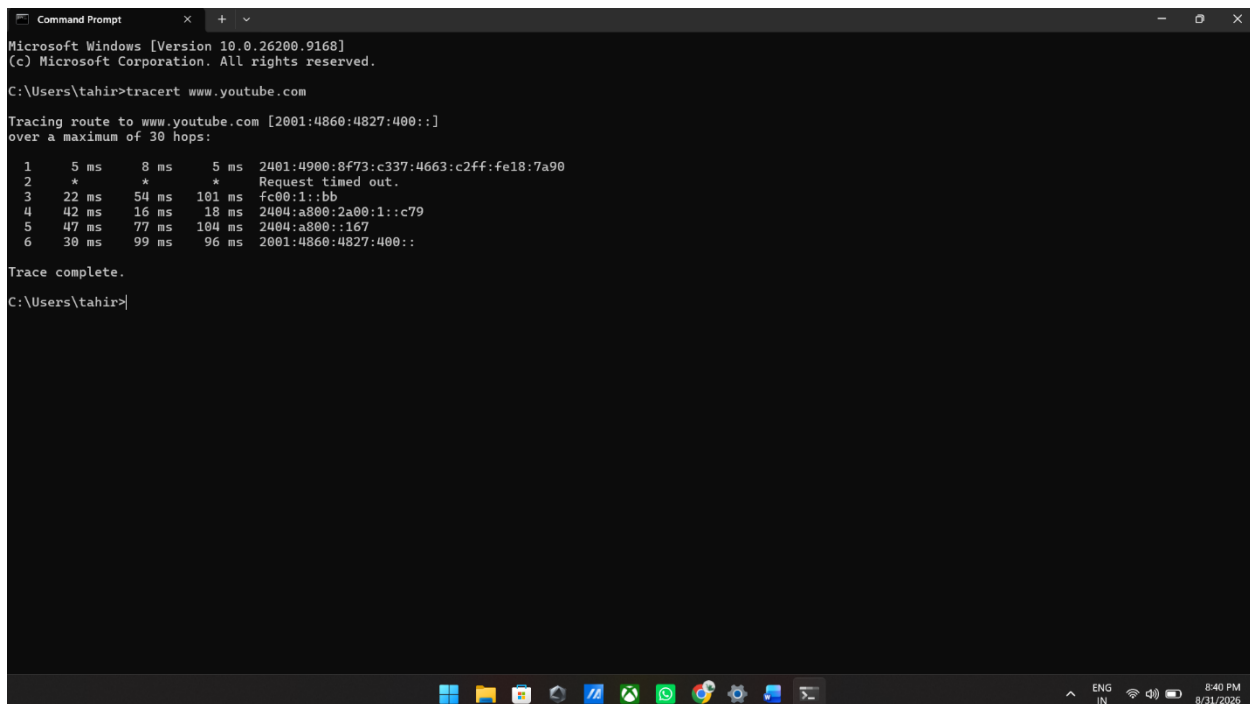
1. Connect the laptop to the **same Wi-Fi** as your friend's phone.
2. Open **Command Prompt (CMD)**.
3. Type `ipconfig` and press **Enter**.
4. Find the laptop's **IPv4 Address**.
5. On the phone, find its **IP address** in Wi-Fi settings.
6. In CMD, type:
`ping PHONE-IP`
7. If you see **Reply from...**, the laptop can communicate with the phone.



Friend's Phone (Android)

1. Connect the phone to the **same Wi-Fi**.
2. Open **Termux** or a ping app.
3. Find the **laptop's IP address** using `ipconfig` on the laptop.
4. In Termux, type:
`ping LAPTOP-IP`
5. If you see **64 bytes from...**, the phone can communicate with the laptop.
6. Press **Ctrl + C** to stop the ping.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.



```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\tahir>tracert www.youtube.com

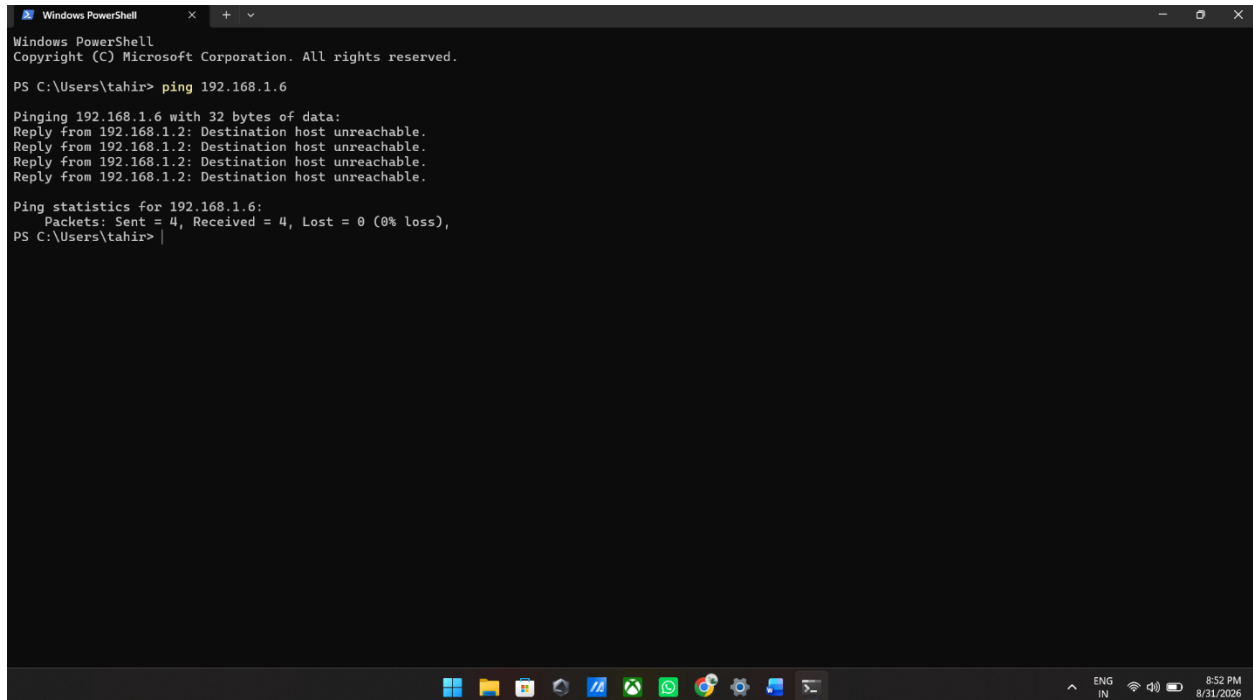
Tracing route to www.youtube.com [2001:4860:4827:400::]
over a maximum of 30 hops:
  0  5 ms   8 ms   5 ms  2401:4900:8f73:c337:4663:c2ff:fe18:7a90
  1  *      *      *      Request timed out.
  2  22 ms  54 ms  101 ms fc00:1::bb
  3  42 ms  16 ms  18 ms  2404:a800:2a00:1::c79
  4  47 ms  77 ms  104 ms 2404:a800::167
  5  30 ms  99 ms  96 ms 2001:4860:4827:400::

Trace complete.

C:\Users\tahir>
```

1. Open **Command Prompt (CMD)**.
2. Type:
tracert www.youtube.com
3. Press **Enter**.
4. Wait for the trace to finish
5. Look at the numbered lines on the left: **1, 2, 3, 4...**
6. The **last numbered line before “Trace complete”** is the number of hops.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.



The screenshot shows a Windows PowerShell window with the following text:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\tahir> ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\tahir> |
```

The taskbar at the bottom shows the Windows Start button, several application icons, and system tray icons including the language (ENG IN), network, and battery status. The date and time are 8:52 PM on 8/31/2026.

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

IP Address	MAC Address	Device
192.168.1.1	44-63-C2-18-7A-90	Wi-Fi Router
192.168.1.2	Your laptop's Wi-Fi adapter	My Laptop
192.168.1.7	98-2C-C6-A3-6A-CF	Other connected device

➤ **My home Wi-Fi has 3 relevant devices shown by arp - a: the router, my laptop, and another device at 192.168.1.7. The device at 192.168.1.7 can be identified by checking the IP address on each phone or gadget's Wi-Fi settings.**

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Hint: Include screenshots or command outputs if possible.

Network Setup and Connectivity Testing Report:

➤ Introduction

I connected my laptop and other devices to the same home Wi-Fi network. The main purpose was to check whether the devices could communicate with each other and to understand how data travels through a network.

➤ Step 1: Connect the Devices

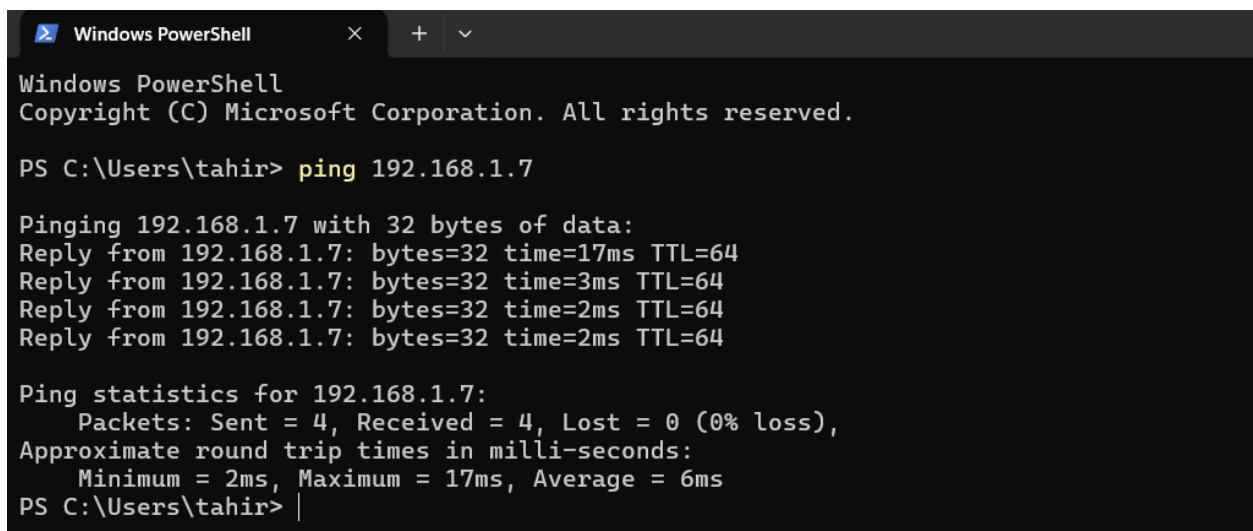
First, I connected my laptop and my friend's phone to the same Wi-Fi network. I checked that both devices were connected to the same router.

➤ Step 2: Check Connectivity Using Ping

I opened Command Prompt on my Windows laptop and used the ping command. I entered the IP address of the other device.

Example:

ping 192.168.1.7

A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text shows the user running the 'ping 192.168.1.7' command. The output displays four successful replies from the target IP, each with 32 bytes of data and varying response times (17ms, 3ms, 2ms, 2ms). It also includes ping statistics: 4 packets sent, 4 received, 0% loss, with minimum, maximum, and average round trip times of 2ms, 17ms, and 6ms respectively.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\tahir> ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:
Reply from 192.168.1.7: bytes=32 time=17ms TTL=64
Reply from 192.168.1.7: bytes=32 time=3ms TTL=64
Reply from 192.168.1.7: bytes=32 time=2ms TTL=64
Reply from 192.168.1.7: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 17ms, Average = 6ms
PS C:\Users\tahir> |
```

The result showed replies from the device, which means the two devices were able to communicate through the Wi-Fi network.

➤ Step 3: Test the Internet Route

Next, I used the `tracert` command to check the route from my laptop to YouTube.

`tracert` www.youtube.com

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\tahir> tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4827:400::]
over a maximum of 30 hops:

  1    1 ms    1 ms    2 ms  2401:4900:8f73:c337:4663:c2ff:fe18:7a90
  2    *      *      *      Request timed out.
  3   73 ms   30 ms  105 ms  fc00:1::bb
  4  123 ms  102 ms  100 ms  2404:a800:2a00:1::c79
  5  117 ms  102 ms  101 ms  2404:a800::167
  6  120 ms   87 ms  114 ms  2001:4860:4827:400::

Trace complete.
PS C:\Users\tahir> |
```

The command displayed several numbered hops. Each hop represents a router or network point that the data passed through before reaching the destination.

➤ Step 4: Check Connected Devices

I also used the `arp -a` command to see devices that my laptop had recently communicated with.

`arp -a`

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\tahir> arp -a

Interface: 192.168.244.1 --- 0x9
Internet Address      Physical Address      Type
192.168.244.254       00-50-56-fc-75-3e     dynamic
192.168.244.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
224.77.77.77          01-00-5e-4d-4d-4d     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

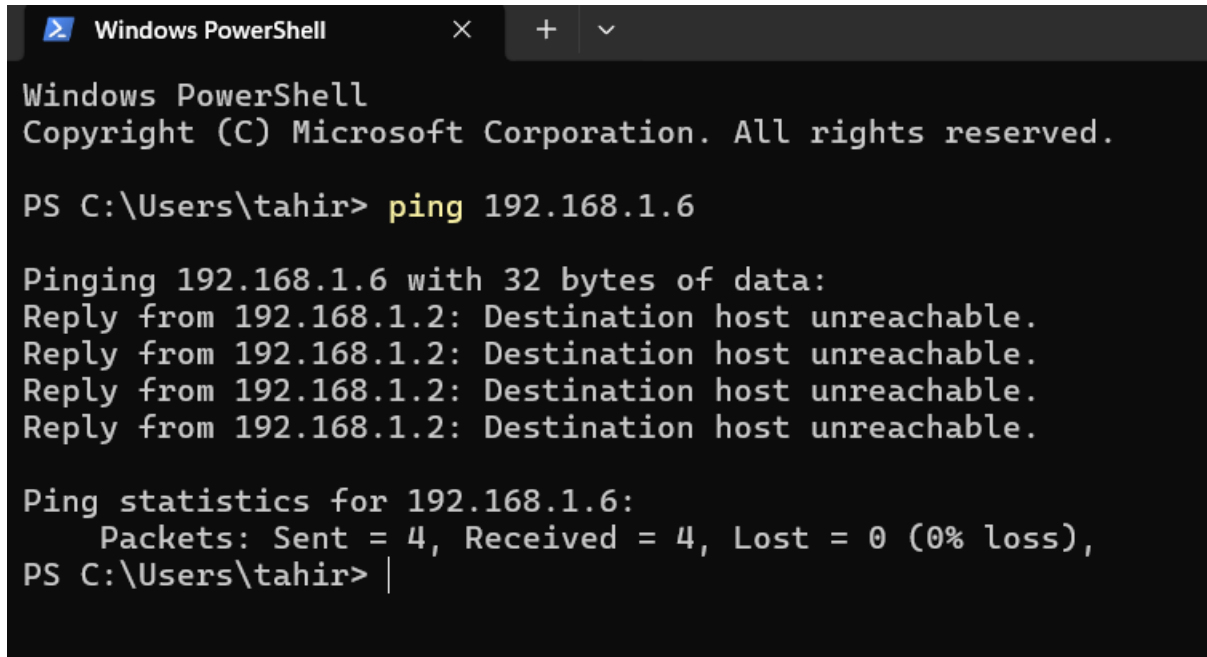
Interface: 192.168.1.2 --- 0xe
Internet Address      Physical Address      Type
192.168.1.1           44-63-c2-18-7a-90     dynamic
192.168.1.7           98-2c-c6-a3-6a-cf     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
224.77.77.77          01-00-5e-4d-4d-4d     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.118.1 --- 0x11
Internet Address      Physical Address      Type
192.168.118.254       00-50-56-e7-09-2b     dynamic
192.168.118.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
224.77.77.77          01-00-5e-4d-4d-4d     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
PS C:\Users\tahir> |
```

I found my router at 192.168.1.1 and another device at 192.168.1.7. My laptop had the IP address 192.168.1.2.

➤ Results and Issues

The ping test was successful when the other device was connected to Wi-Fi. When the device was disconnected, the ping test could not receive a reply.

A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text shows the user running 'ping 192.168.1.6'. The output indicates that the destination host is unreachable for all four attempts. The ping statistics show 4 packets sent, 4 received, and 0% loss, which is unusual given the 'unreachable' status.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\tahir> ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Users\tahir> |
```

The tracert command successfully showed the route to YouTube. The arp -a command also showed the router and another device on the local network.

➤ Conclusion

This activity helped me understand how devices communicate over Wi-Fi. I learned how to use ping to test connectivity, tracert to see the path to a website, and arp -a to view nearby network devices. Overall, the network worked properly and the tests were successful.